Modeling Wi-Fi Traffic in Hot-Spots & Video over Wireless (demo)

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Princeton University SCENIC

Social and Clean Edge Network via Inter-disciplinary Collaboration

The Princeton EDGE Lab

Modeling Wi-Fi Traffic in Hot-Spots

Outline

- Overview of Data
- Arrival Count Modeling
- Connection Duration Modeling
- Simultaneous Users Modeling

Motivation

Free WiFi Coming to Starbucks, Now a Seat Will be Harder to Find

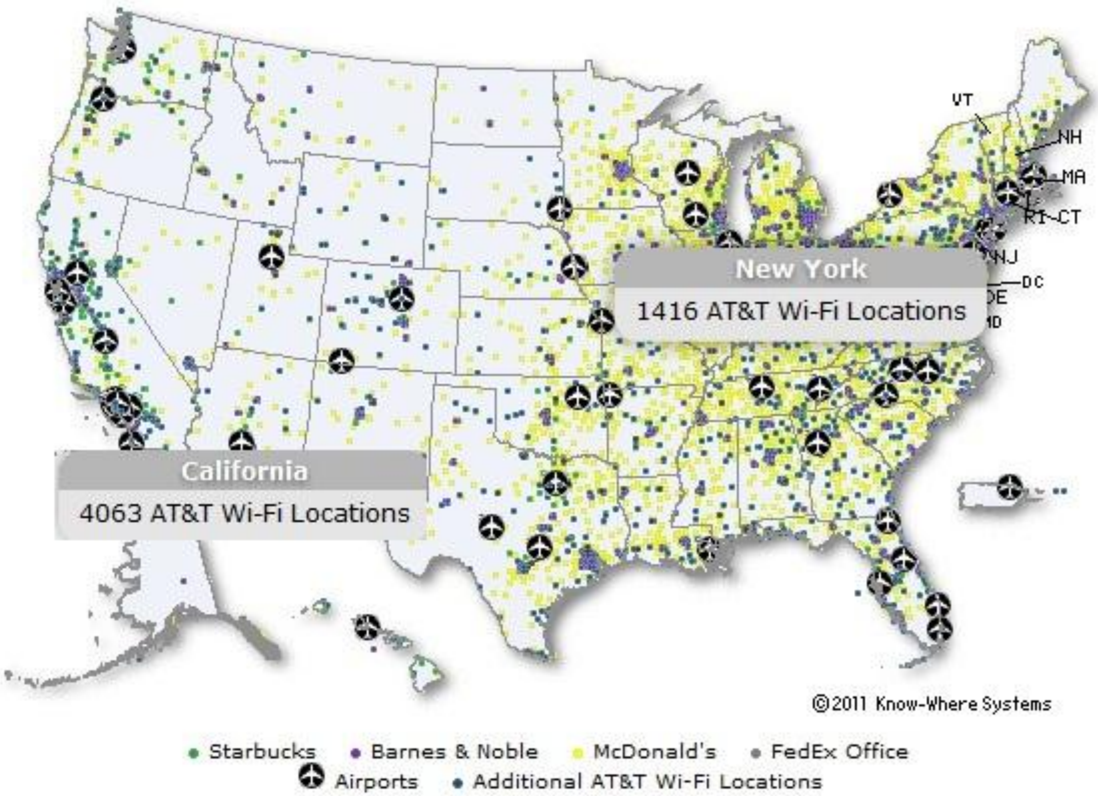
Source: AP • June 14, 2010 6 retweet Share

In an effort to increase sales and bring in more customers, Starbucks will now be offering free WiFi service as of July

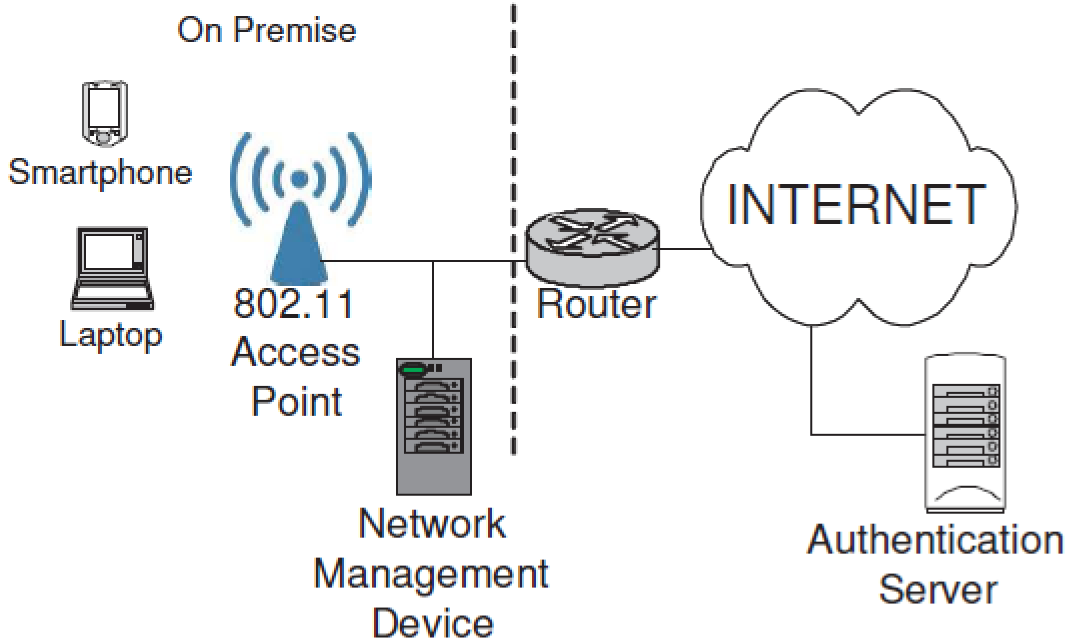
ır using the wireless Internet in Starbucks stores while mobile phone ıutes on the network.

Seattle chain were charged $3.99 for two additional hours.

ıe to be offered through AT&T. But it won't require a Starbucks loyalty y by CEO Howard Schultz, who spoke at a conference in New York.



Data Collection



Mobile Internet Access using Wi-Fi Hotspots

Overview of Data

- Wi-Fi data collected by AT&T in March 2010 in New York and San Francisco
  - Coffee shops, fast food chains, book stores, hotels, …
  - Attributes:
    - Connection login/logout times
    - Bytes uploaded/downloaded
    - Venue size (small, medium, large), zip codes, …

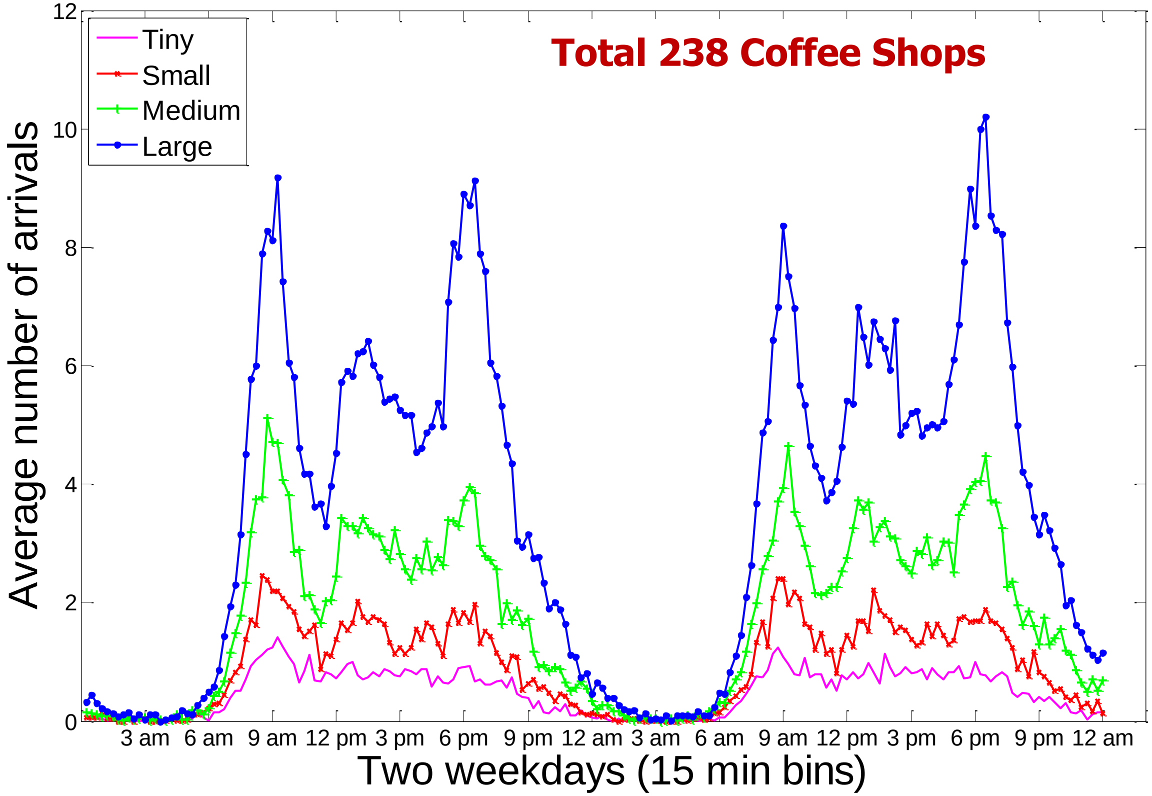
# of customers	**234,742**
# of devices	10
# of connections	1,322,541
# of cities	2 (NYC, SF)
# of Wi-Fi venues	362
# of zip codes	87
Trace duration	4 weeks

Goals

- Realistic modeling of
  - Session arrivals
  - Connection duration distribution
  - Number of simultaneously present customer distribution

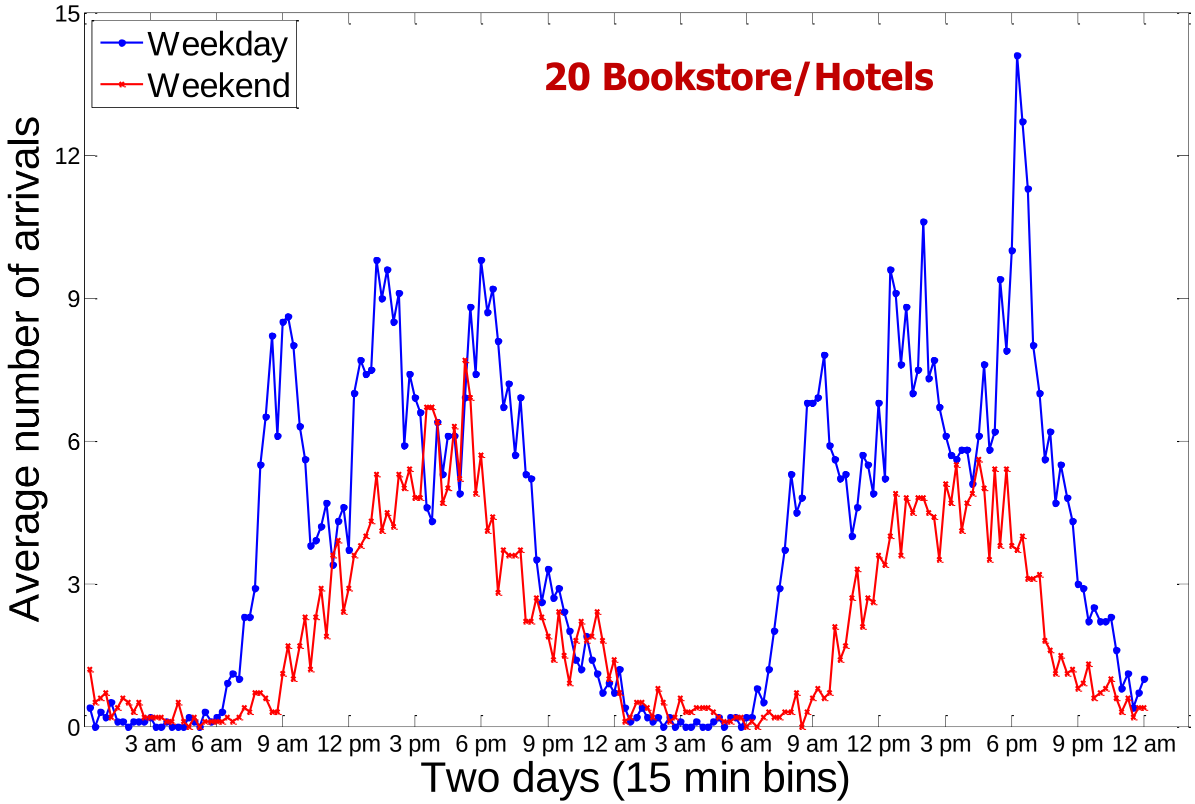
Network Capacity Planning

Arrival Trends



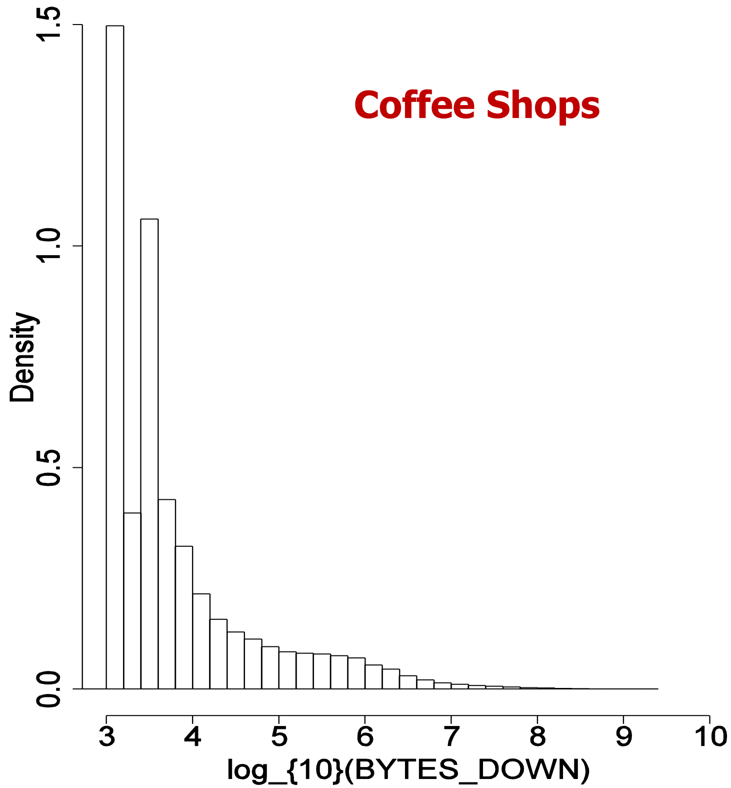
- Arrival rates vary drastically within the same business type
- Characteristic peaks in means across all categories within same business type

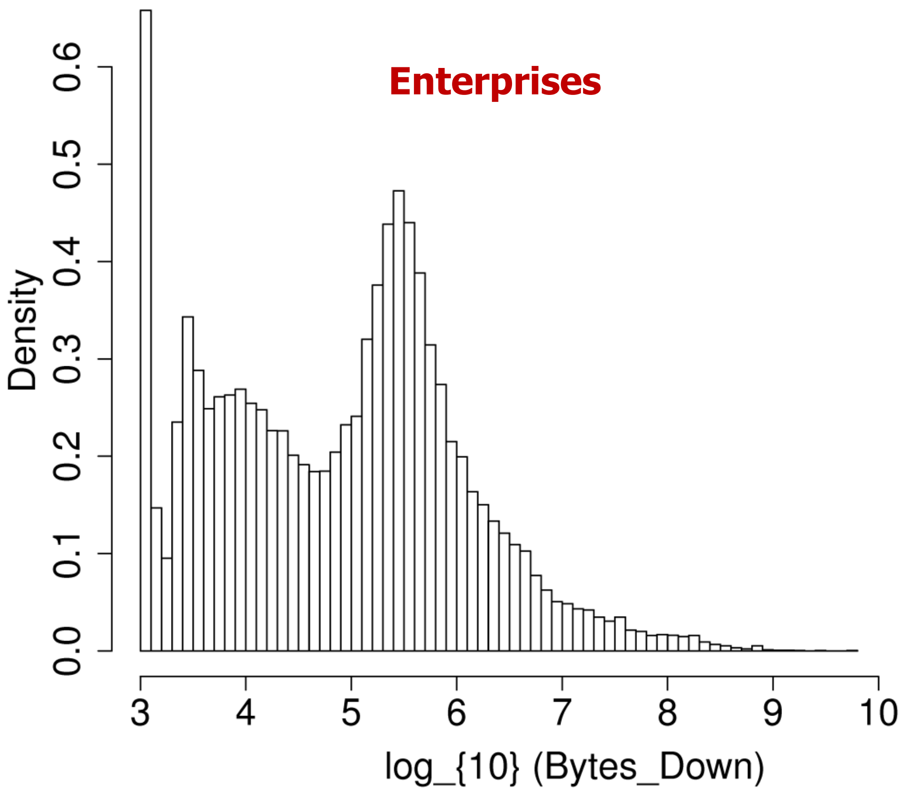
Arrival Trends



- Significantly different weekday and weekend patterns

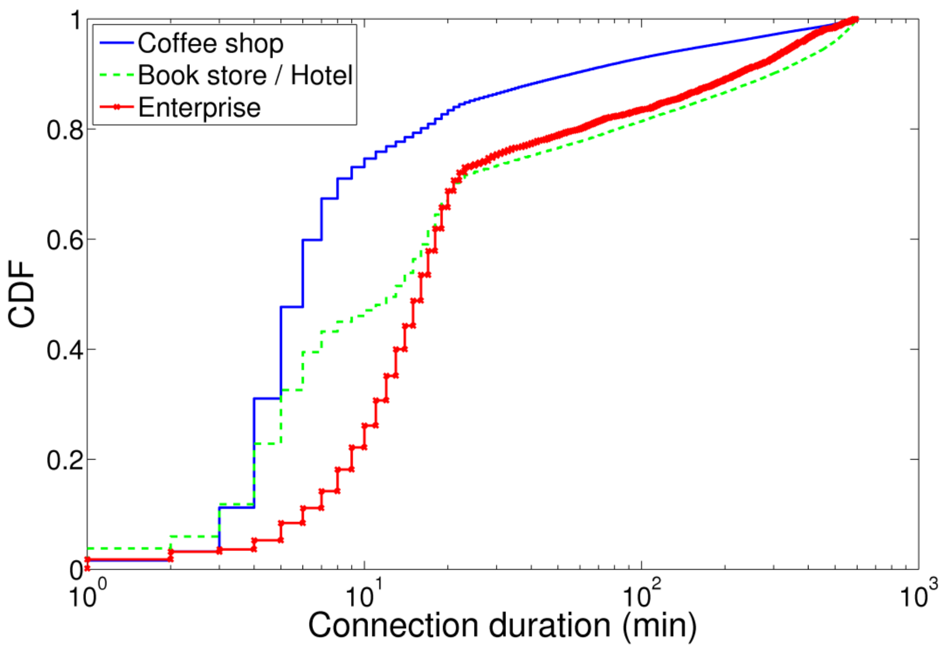
Byte Counts



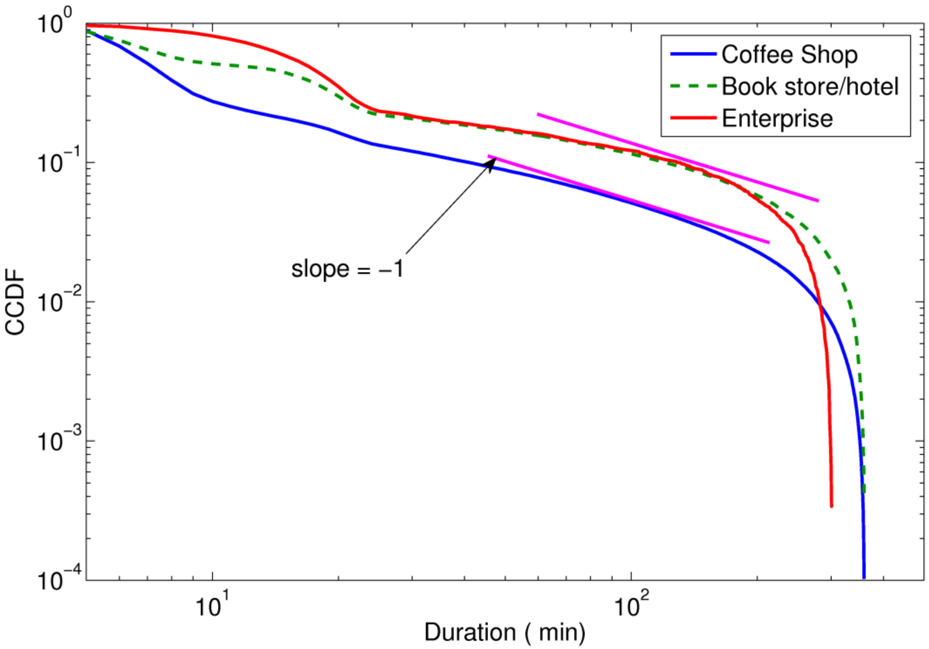


- Coffee shops: typically a few KB
- Enterprises: typically a few MB to a few GB
- Long tails

Connection Durations



CDF by business types



Complimentary CDF (log-log scale)
=> Long tails

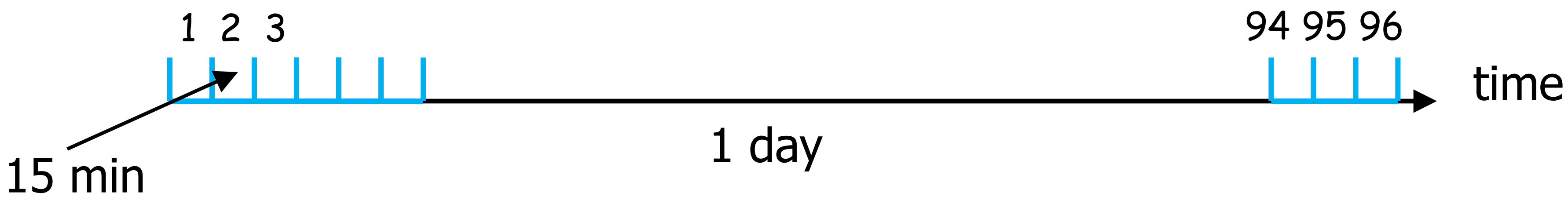
Connection Duration (min)	Mean	S.D.
Coffee shops & fast food chains	29.8	81.9
Book stores & hotels	73.4	142.3
Enterprises & stadiums	61.6	113.8

Arrival Count Modeling

- Data showed time-dependent arrival rates
  - MMPP fails
    - Models arrival counts with constant periods of arrival rate
  - Polynomial curve fitting to the observed mean
    - Poor performance
    - Could not capture within-day pattern with small no. of terms
  - Standard Poisson regression fails

- Non-homogeneous Poisson regression with clustering

Arrival Count Modeling



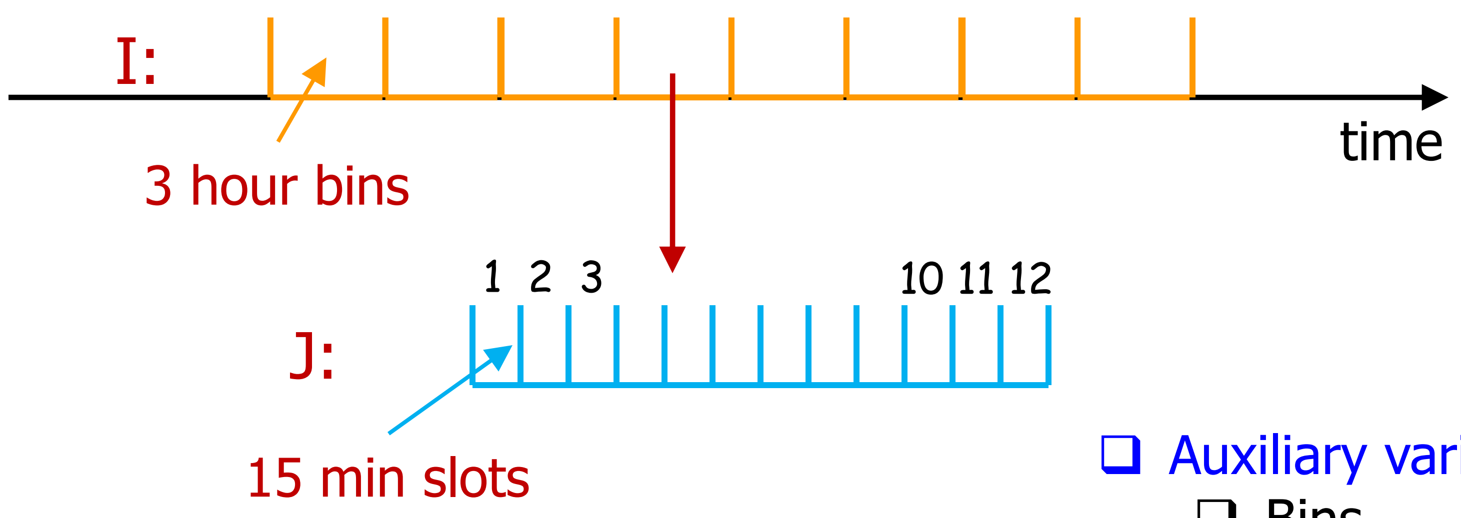
- K-Means Clustering
  - Average number of arrivals do not differ much within each group
  - Automatic 24 hour wrap-around in clustering

Cluster	Slots
0	1–28, 94–96
1	83–87
2	32–34, 38–40, 64–66
3	35–37, 72–75
4	48–58
5	41–47, 59–63, 67–71, 76–78
6	79–82
7	29–31, 88–93

- Clusters of 15 min time slots over a day
- Non-contiguous busy slots (35-37, 72-75) map to a common cluster

Arrival Count Modeling

- Non-stationary Poisson Process
  - Time-dependent deterministic arrival rate
  - Divide time into 3 hour bins I: 8 bins per day
  - Divide each bin into 15 min slots J: 12 slots per bin



- Auxiliary variables:
  - Bins
  - Slots

Arrival Count Modeling

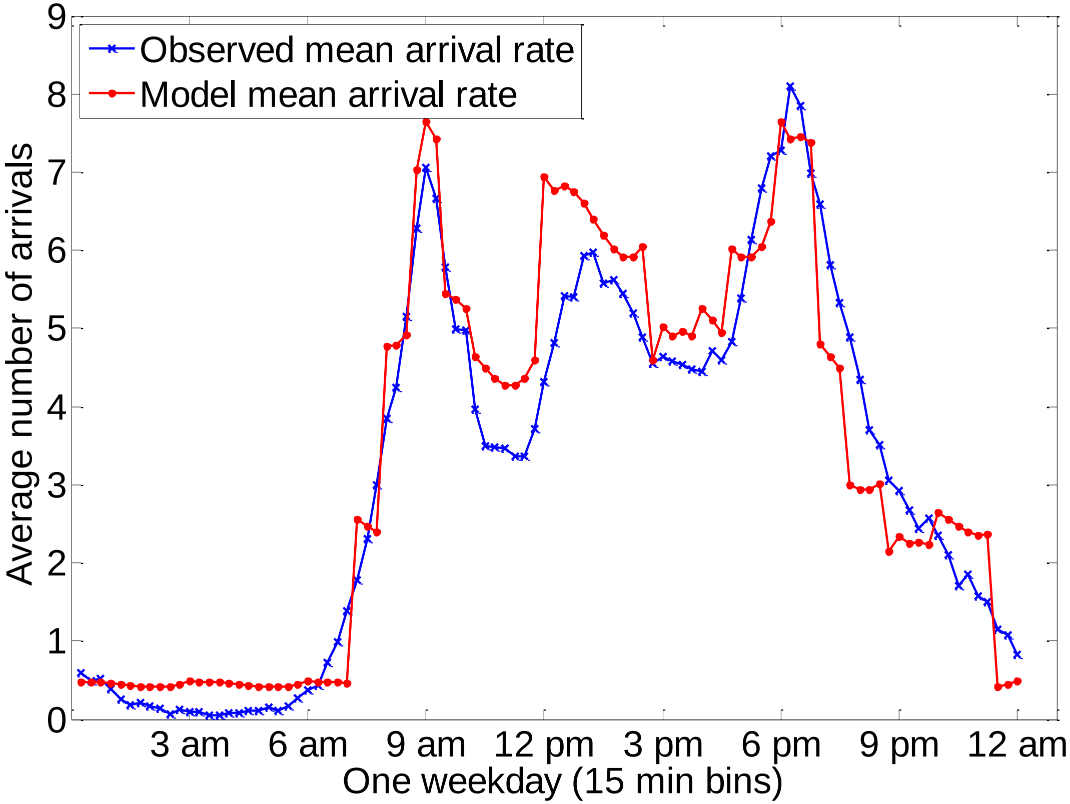
- Poisson Regression Model (GLM)
  - Polynomial type dependence on bin and slot numbers

$$\log(\lambda(I,J)) = \alpha + \sum_{k=1}^{3} \beta_k I^k + \sum_{k=1}^{3} \gamma_k J^k + \delta(I * J)$$

- First term: Over-a-day mean behavior
- Sum terms: Differential effects of specific cluster and slots within it
- Last term: (Interaction term) – differential effect of slot J does not have to be the same across all clusters

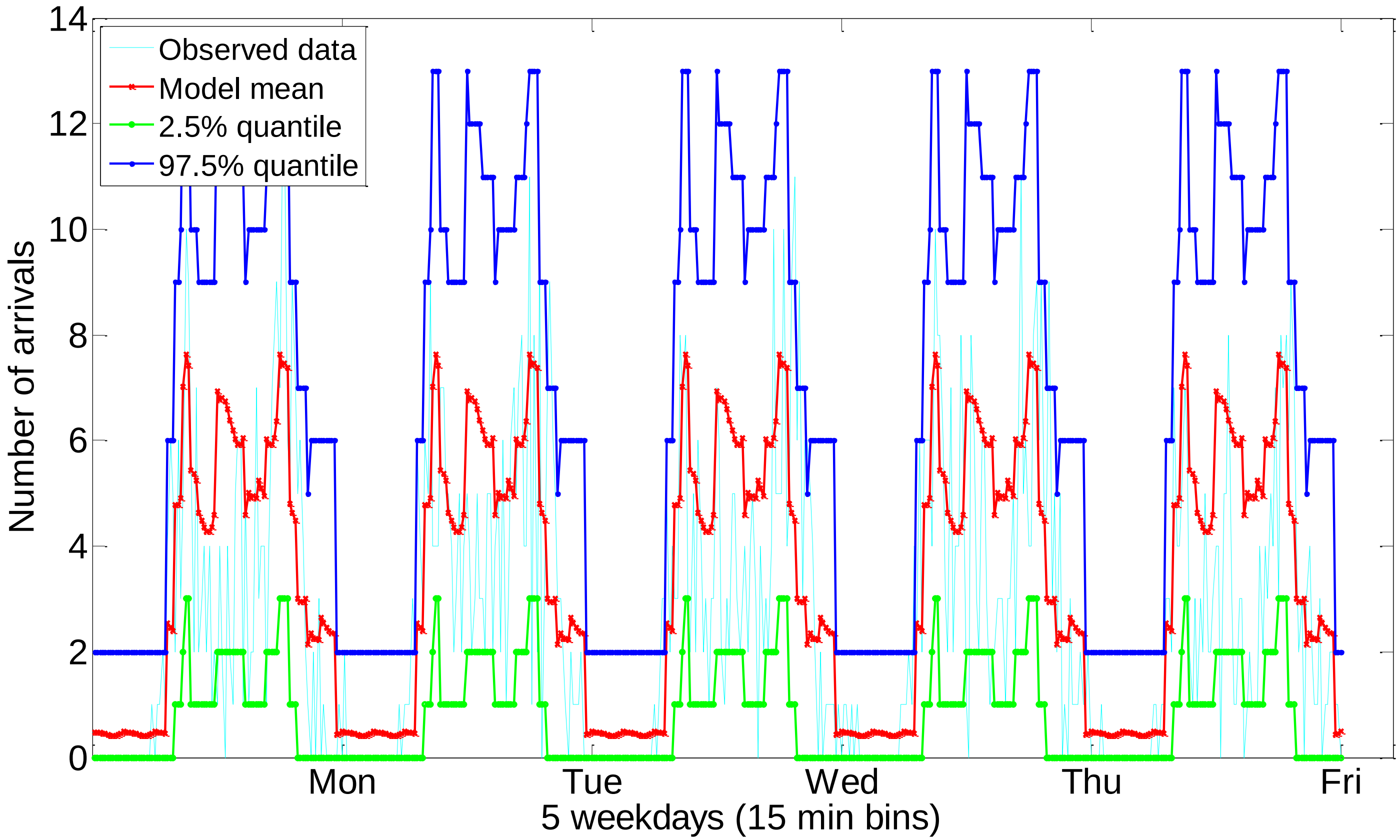
Results: Arrivals

- Training data (3 weeks)
  - 649,501 arrivals
- Test data (1 week)
  - 225,085 arrivals



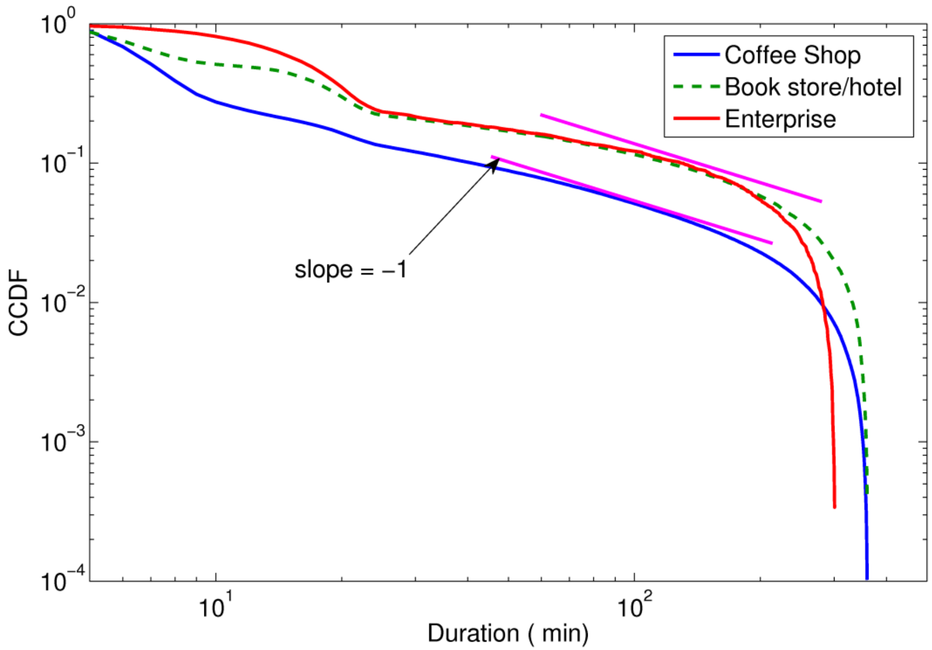
- **Coffee shops:** Observed mean arrival rate plotted against the model mean arrival rate; these provide intra--day patterns for a cluster by averaging over its members

Results: Arrivals



❑ **Coffee Shops:** Model mean arrival rate along with the 97.5% quantile and 2.5% quantile bands plotted against 5 days of validation data for an example coffee shop.

Session Duration Modeling



- ❑ Few seconds to several hours with a very long tail
- ❑ Sizeable mass at head: 78% is at most 10 min
- ❑ Need a distribution that matches the entire range: from head to tail

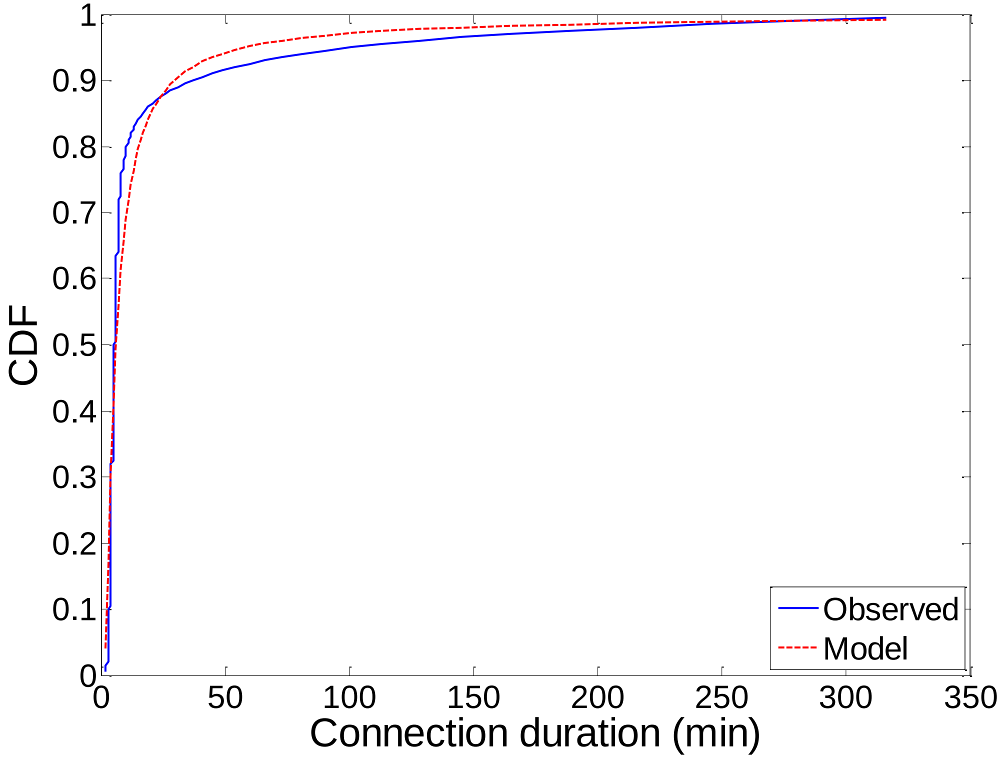
- Model the logarithm of duration (Y) as a Phase-Type (PH) random variable (X)

$$Y \sim e^X$$

PH-Type Distribution

- PH-type random variable
  - Sum of a random number of exponential r.v.s
  - Distribution time to absorption in a Markov Process
  - Dense in the class of all distributions
    - Captures both tails and heads, as opposed to Pareto and Weibull
  - Exponentially decaying tail asymptotically going to 0 as $e^{-\eta x}$ where η is the real Eigen value of the rate transition matrix

Results: Duration



- ❑ **Coffee Shops:** CDF plot of durations for coffee shops and data (truncated at 6 hours)

- ❑ Phase type distributions were fit using the EM algorithm

- ❑ A fit of order 5 was found to be adequate

Simultaneous Connections

- Arrivals
  - Non-homogeneous Poisson process (time-dependent, deterministic arrival rates)
- Connection Durations
  - PH-type distribution
- Simultaneous number of connections
  - Number of busy servers in a Queuing model

$$M_t/G/\infty$$

Simultaneous Connections

- Theorem

  The number of busy servers $Q(t)$ (i.e., number of simultaneous connections), at time t follows a Poisson distribution with mean $m(t)$, given by:

$$m(t) = \int_0^t \lambda(\tau)[1 - H(t-\tau)]\, d\tau$$

  where H() is the service time distribution

Simultaneous Connections

- Novel proof based on semi-regenerative argument
  - Does not require the system to be empty at some infinite past
  - Simple, transparent, and general

- Shows that the Probability Generating Function G(t) of Q(t) is

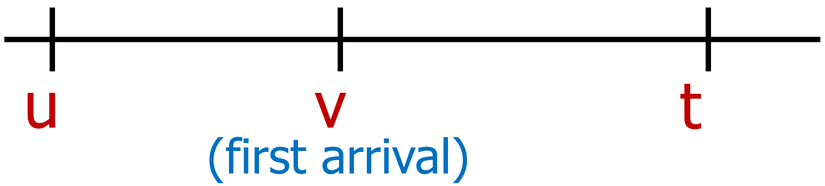
Power series representation of the pmf (discrete r.v.s)

For Poisson r.v.s

$$
\begin{aligned}
G_t(z,0) &= E[z^{Q(0,t)}] \\
&= e^{(z-1)\int_0^t \lambda(\tau)\bar{H}(t-\tau)\,d\tau}
\end{aligned}
$$

Simultaneous Connections

- Proof idea (embed into a larger problem):



  - Q(u,t): number of customers who arrive in (u,t] and are still there at t
  - No arrivals in (u,t]
  - First arrival occurs at some v in (u,t]
    - The arrival leaves before t
    - The arrival still remains at t

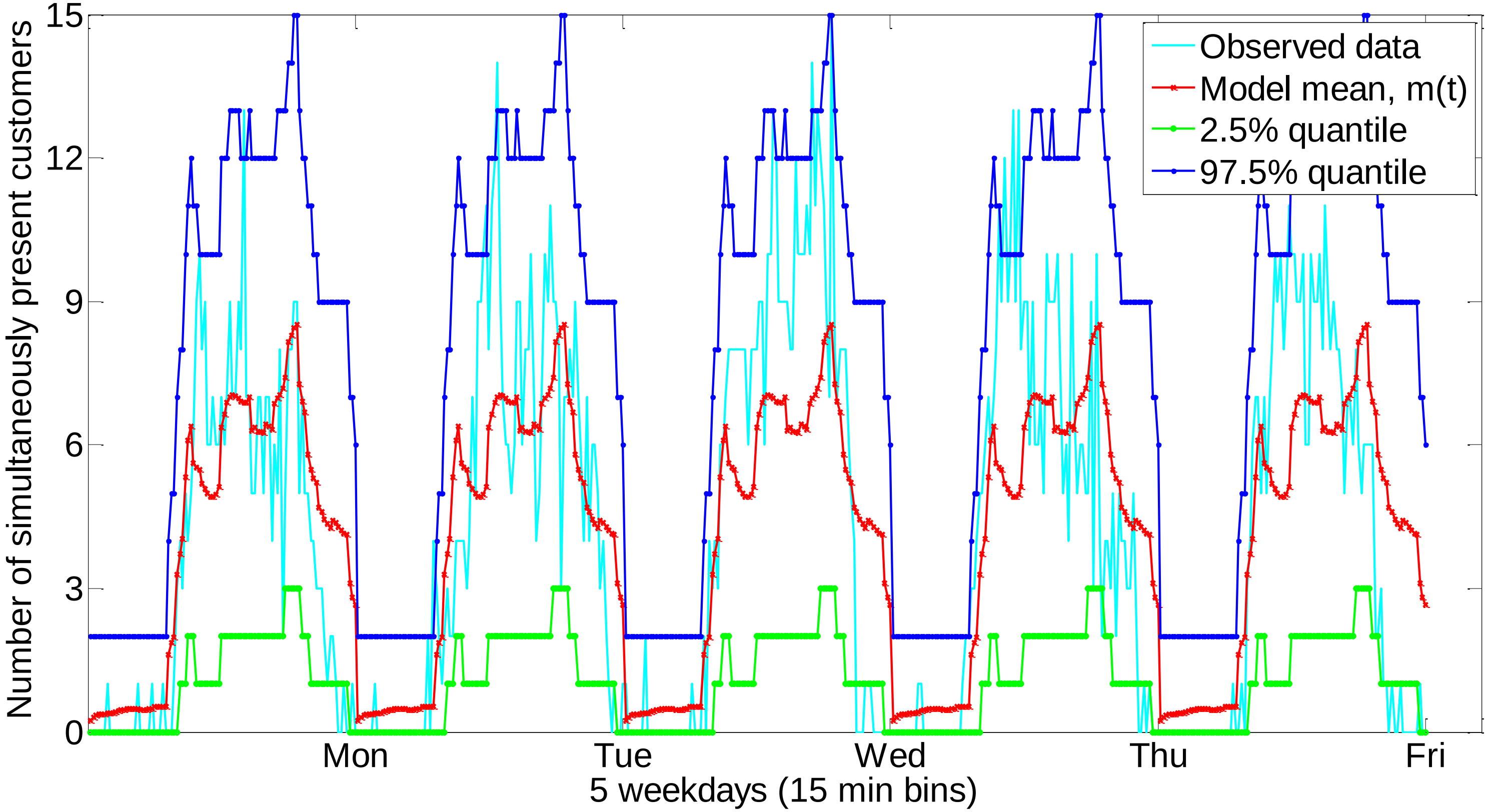
Simultaneous Connections

- Solve the integral equation

where $\Lambda(t) = \int_0^t \lambda(\tau)\, d\tau$

$$
\begin{aligned}
G_t(z,u) &= E[z^{Q(u,t)}] \\
&= \boxed{e^{-[\Lambda(t)-\Lambda(u)]}} + \\
&\quad \int_u^t \boxed{\lambda(v) e^{-[\Lambda(v)-\Lambda(u)]}} \cdot \\
&\quad \boxed{\{H(t-v) + z[1 - H(t-v)]\}} G_t(z,v)\, dv
\end{aligned}
$$

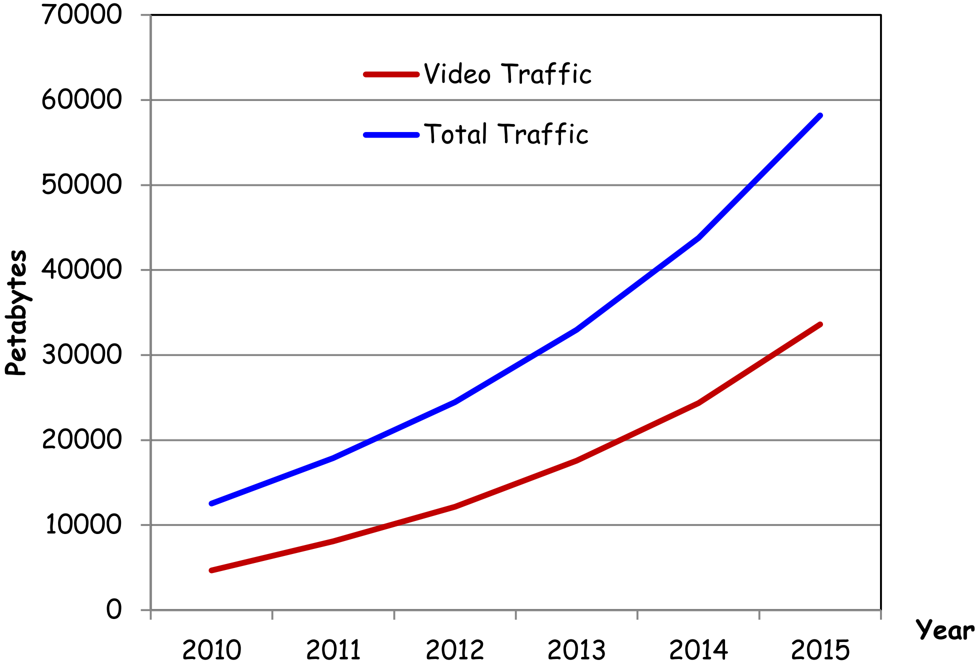
Results: Simultaneous Connections



❑ **Coffee Shops:** Expected number of simultaneously present customers along with the **97.5% quantile** and **2.5% quantile** bands plotted against 5 days of validation data for an example coffee shop.

Video over Wireless (demo)

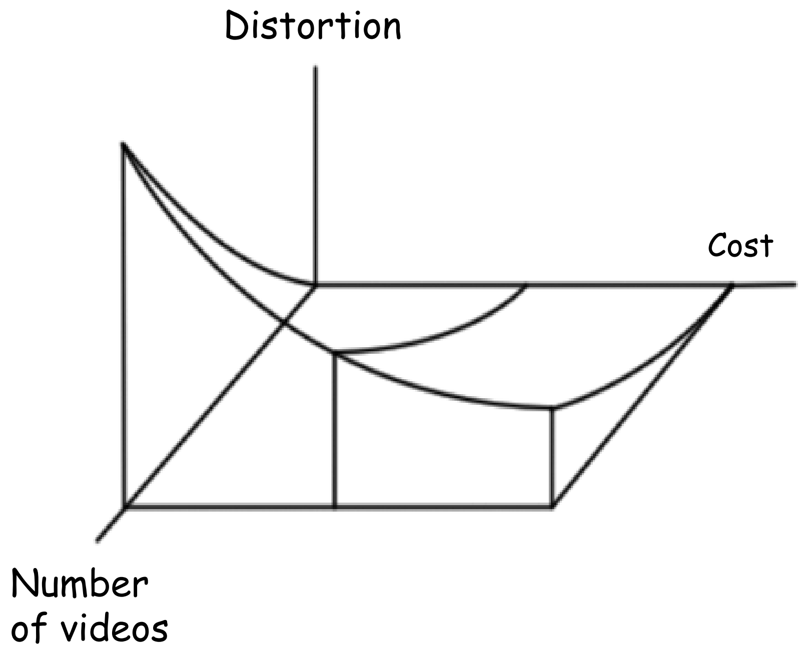
Conflicting Market Trends



30 vs. **10**

- **30**: % of downstream Internet traffic from Netflix during peak hours in the US
- **10**: $$ per GB charged by AT&T and Verizon wireless for data usage above 2 GB

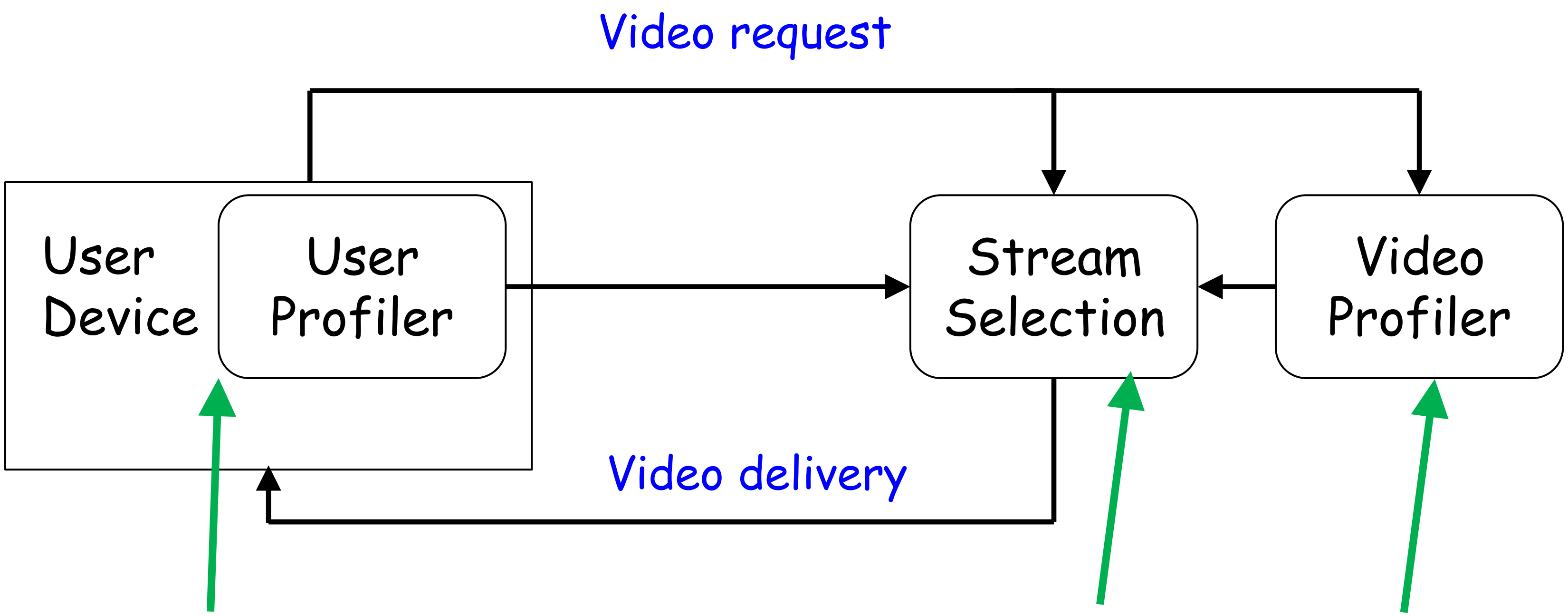
3-Dimensional Trade-off



Question

Is there a way for the consumer to stay within her monthly quota and watch videos without suffering noticeable distortion ?

QAVA: Quota Aware Video Adaptation



Prediction of user **consumption pattern** from past usage by online learning

Adaptively choose **bit rates** to deliver to user

Estimate video **compressibility** from motion vectors

Conclusions

- A modeling framework for Wi-Fi traffic in large-scale public hotspots

Capacity Planning

  - Arrival count modeling using statistical clustering and non-stationary Poisson model
  - Use of Phase-Type r.v. to model the logarithm of long-tailed durations
  - Simultaneously present customer modeling using a $M_t/G/\infty$ queuing model

- Novel proof on semi-regenerative argument for the number of busy servers

- A practical, end-to-end, quota-aware video delivering system exploiting video compressibility

- Amitabha Ghosh, Rittwik Jana, V. Ramaswami, Jim Rowland, and N. K. Shankaranarayanan, Modeling and Characterization of Large-Scale Wi-Fi Traffic in Public Hot-Spots, *INFOCOM 2011*, Shanghai, China, April 2011.

- Jiasi Chen, Amitabha Ghosh, Mung Chiang, QAVA: Quota Aware Video Adaptation, (under submission)

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Thank you!